UNIT-2 AGILE MINI-PROJECT

Student Record and Academic Management Platform

Agile Project Planning & Scrum Documentation

Prepared as per the Unit-2 Agile Mini-Project activities and deliverables.

1. Study of Business Problem and Product Vision

The project is a Student Record and Academic Management Platform. The system is intended to help educational institutions manage student records, attendance, academic performance, course registration and reporting through a scalable web-based application.

Key areas reviewed: Product Vision, Mission Statement, stakeholders, business requirements, functional and non-functional requirements, user personas, Minimum Viable Product (MVP), and product roadmap.

Stakeholders: Students, faculty members, administrators and institutional management.

Business requirements: Maintain accurate student information, manage courses and attendance, record academic performance, provide secure access, and generate useful reports.

Functional requirements: Registration, profile management, student search/update/delete, course management, attendance, marks, authentication, notifications and reporting.

Non-functional requirements: Security, usability, reliability, scalability, maintainability and acceptable performance.

2. Agile Product Vision

Product Vision: Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

Overall objective: Deliver a simple, secure and maintainable platform that reduces manual academic-record work and improves access to accurate information.

Primary users: Students, faculty members and administrators.

Business benefits: Faster record management, improved data accuracy, easier attendance/performance tracking, centralized information and better reporting.

Success criteria: Core user journeys work correctly; records can be created, updated and searched; attendance and marks are reportable; authentication is secure; and sprint deliverables meet their acceptance criteria.

Constraints: Limited development time, team capacity, implementation complexity and dependency on reliable data/storage.

3. Scrum Team Structure and Roles

Role	Responsibilities
Product Owner	Defines requirements, prioritizes Product Backlog, represents stakeholder interests and accepts completed
Scrum Master	Facilitates Scrum ceremonies, removes impediments, supports collaboration and ensures Agile practices ar
Development Team	Designs the application, writes code, tests software, prepares documentation and demonstrates completed

Suggested team allocation: one Product Owner, one Scrum Master, and the remaining members as the Development Team. For an individual/team submission, responsibilities may be shared as agreed by the team.

4. Project Epics

Epic	Scope
Epic 1 – Student Management	Student Registration; Student Profile Management; Student Search; Student Update; Student Deletion.
Epic 2 – Course Management	Add Course; Update Course; Assign Students; View Course Details.
Epic 3 – Attendance Management	Record Attendance; View Attendance; Attendance Reports.
Epic 4 – Marks and Academic Performance	Enter Marks; Update Marks; Calculate Grades; Generate Result Sheets.
Epic 5 – Authentication and Security	User Login; Logout; Password Reset; Role-Based Access Control.
Epic 6 – Notifications	Attendance Alerts; Examination Notifications; Result Notifications.

Epic	Scope
Epic 7 – Reporting	Student Reports; Attendance Reports; Performance Reports; Course Reports.

5. User Story Repository & Acceptance Criteria

US-01 – Authentication

User Story: As a registered user, I want to log in securely so that I can access the appropriate dashboard.

Acceptance Criteria: Login page loads; registered users can log in; invalid credentials show an error; successful login redirects to dashboard; session management works.

Priority: High **Story Points:** 5

US-02 – Student Management

User Story: As an administrator, I want to add a student so that the student's academic record can be maintained.

Acceptance Criteria: Required fields are validated; student is saved; unique student ID is enforced; success confirmation is shown.

Priority: High **Story Points:** 5

US-03 – Student Management

User Story: As a user, I want to search for a student so that I can quickly find records.

Acceptance Criteria: Search accepts relevant student information; matching records appear; no-match state is shown clearly.

Priority: High **Story Points:** 3

US-04 – Student Management

User Story: As an administrator, I want to update student details so that records remain current.

Acceptance Criteria: Existing record loads; edits are validated; changes are saved; updated data is displayed.

Priority: High **Story Points:** 3

US-05 – Student Management

User Story: As an administrator, I want to delete a student so that obsolete records can be removed appropriately.

Acceptance Criteria: Correct record is selected; confirmation is required; record is removed; user receives confirmation.

Priority: Medium **Story Points:** 5

US-06 – Course Management

User Story: As an administrator, I want to add and update courses so that course information remains accurate.

Acceptance Criteria: Course details validate; course can be saved and edited; course details are viewable.

Priority: Medium **Story Points:** 5

US-07 – Attendance

User Story: As a faculty member, I want to record attendance so that attendance records remain accurate and updated.

Acceptance Criteria: Faculty selects course and lecture date; attendance can be marked; records save successfully; reports can be generated.

Priority: High **Story Points:** 5

US-08 – Marks

User Story: As a faculty member, I want to enter marks so that student performance can be recorded.

Acceptance Criteria: Student/course can be selected; marks validate; data saves; grades/results can be generated.

Priority: High **Story Points:** 5

US-09 – Notifications

User Story: As a student, I want to receive academic notifications so that I do not miss important updates.

Acceptance Criteria: Notifications are generated for configured events; user can view them; relevant message is displayed.

Priority: Medium **Story Points:** 3

US-10 – Reporting

User Story: As an administrator, I want reports so that I can monitor students, attendance, performance and courses.

Acceptance Criteria: Report type can be selected; accurate data is shown; report can be generated successfully.

Priority: Medium **Story Points:** 5

6. Prioritized Product Backlog

Priority	ID	Epic	User Story	Points
High	US-01	Authentication	As a registered user, I want to log in securely so that I can access the appropriate dashboard.	5
High	US-02	Student Management	As an administrator, I want to add a student so that the student's academic record can be maintained.	5
High	US-03	Student Management	As a user, I want to search for a student so that I can quickly find records.	3
High	US-04	Student Management	As an administrator, I want to update student details so that records remain current.	3
Medium	US-05	Student Management	As an administrator, I want to delete a student so that obsolete records can be removed appropriately.	5
Medium	US-06	Course Management	As an administrator, I want to add and update courses so that course information remains accurate.	5
High	US-07	Attendance	As a faculty member, I want to record attendance so that attendance records remain accurate and upd	5
High	US-08	Marks	As a faculty member, I want to enter marks so that student performance can be recorded.	5
Medium	US-09	Notifications	As a student, I want to receive academic notifications so that I do not miss important updates.	3
Medium	US-10	Reporting	As an administrator, I want reports so that I can monitor students, attendance, performance and course	5

Prioritization is based on business value, user importance, technical dependencies and implementation complexity.

7. Story Point Estimation

Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

The team can use Planning Poker to collaboratively estimate effort.

8. Sprint Planning

Sprint 1

Goal: Authentication and Project Setup

Activities: Repository initialization; user registration; login functionality; basic user interface; testing.

Expected deliverable: Working authentication flow and basic UI.

Sprint 2

Goal: Student Management Module

Activities: Add student; search student; update student; delete student.

Expected deliverable: Working student management module.

Sprint 3

Goal: Course and Attendance Management

Activities: Add course; assign courses; record attendance; attendance reports.

Expected deliverable: Working course and attendance features.

9. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Course	Add Student	Search Student	Reports	Setup
Notifications	Attendance	Login	Attendance Reports	
Marks	Update Student			

The Scrum Board should be updated throughout sprint execution to make work status visible.

10. Scrum Ceremonies

Sprint Planning: Select User Stories, estimate effort and define the Sprint Goal.

Daily Scrum: Each member answers: What did I complete yesterday? What will I do today? What challenges am I facing?

Sprint Review: Demonstrate completed features and collect stakeholder/team feedback.

Sprint Retrospective: Discuss what went well, challenges encountered, and improvements for the next sprint.

11. Sprint Review Report

At the end of each sprint, the team demonstrates the completed increment against the acceptance criteria. Feedback is recorded as backlog items where appropriate. The review confirms which stories are complete and identifies work that should continue in a future sprint.

12. Sprint Retrospective Report

What went well: Clear user stories, prioritized work, incremental delivery and regular communication.

Challenges: Estimation uncertainty, technical dependencies, testing effort and changing priorities.

Improvements: Refine stories earlier, improve task breakdown, test continuously and make impediments visible quickly.

13. Burndown Chart Data

Sprint	Day	Remaining Story Points
Sprint 1	0	15
Sprint 1	1	12
Sprint 1	2	8
Sprint 1	3	4
Sprint 1	4	0
Sprint 1	5	0
Sprint 2	0	18
Sprint 2	1	15

Sprint	Day	Remaining Story Points
Sprint 2	2	11
Sprint 2	3	7
Sprint 2	4	3
Sprint 2	5	0
Sprint 3	0	18
Sprint 3	1	15
Sprint 3	2	11
Sprint 3	3	7
Sprint 3	4	3
Sprint 3	5	0

The above is a sample planned burndown dataset for tracking sprint progress; actual remaining points should be updated from the team's real sprint board.

14. Agile Framework Comparison

Framework	Main Focus	Suitability
Scrum	Time-boxed sprints, defined roles and ceremonies.	Highly suitable for this student project because it supports structured incr
Kanban	Continuous flow and visual work management.	Useful for continuous maintenance or support work.
XP	Engineering practices and frequent feedback.	Useful when strong coding/testing practices are a major priority.
Lean Development	Eliminating waste and maximizing value.	Useful for improving efficiency and reducing unnecessary work.
LeSS	Scaling Scrum across multiple teams.	More appropriate for larger products with multiple Scrum teams.
SAFe	Structured scaling of Agile across large organizations.	Better suited to large, complex enterprise environments.
Scrum@Scale	Scaling Scrum using coordinated Scrum teams.	Useful when several Scrum teams must coordinate.

Recommended framework: Scrum, because the Unit-2 project is naturally organized into a Product Backlog, short development Sprints, Scrum roles, ceremonies and incremental deliverables.

15. Project Timeline and Release Roadmap

Phase	Planned Work	Release Output
Sprint 0 / Setup	Requirements review, backlog setup, repository and architecture planning	Project setup
Sprint 1	Authentication and basic UI.	Authentication increment
Sprint 2	Student Management.	Student management increment
Sprint 3	Course and Attendance Management.	Course + attendance increment
Future Release	Marks, notifications, reporting, refinements and additional analytics	Expanded platform

16. Complete Agile Documentation Checklist

✓ Agile Project Planning Report

✓ Product Vision Document

✓ Scrum Team Structure

✓ Scrum Roles and Responsibilities Document

✓ Epic Documentation

✓ User Story Repository

✓ Acceptance Criteria Document

✓ Prioritized Product Backlog

✓ Story Point Estimation Sheet

✓ Sprint Planning Document

✓ Sprint Backlog

✓ Scrum Board

✓ Burndown Chart

✓ Sprint Review Report

✓ Sprint Retrospective Report

✓ Agile Framework Comparison Report

✓ Project Timeline and Release Roadmap

Conclusion

This Unit-2 Agile Mini-Project converts the Student Record and Academic Management Platform into an Agile delivery plan. The project is decomposed into epics and user stories, acceptance criteria are defined, work is prioritized and estimated, and the backlog is organized into sprints. Scrum is recommended as the working framework for incremental development, review and continuous improvement.